Clone the Navigator

An unplugged behavior cloning activity · about 45 minutes · instructor guide

The idea. One student, the **Expert**, walks through the crowded room to a goal. A **Scribe** writes down, at every step, what the robot's three sensors would read and what the Expert did. The class turns those cards into a rule (the policy). A second student, the **Robot**, follows that rule and nothing else. Does it get across?

Set up

- **Room:** tables and chairs are the obstacles. Tape a **start** and a **goal**. Two or three different starts work best. Tape a floor grid if you can, so one “step” means the same thing to everyone.
- **Materials:** the scribe worksheet (one per scribe), a second ink color, a timer, a blindfold or sunglasses for the Robot (optional).

The robot’s senses and actions

Sensors (yes/no, one step away)	Actions (one per step)
F front blocked? **L** left blocked? **R** right blocked? That is 8 possible readings. **Extension:** add the goal direction (ahead / left / right / behind): 32 readings.	**Forward:** step ahead. **Left:** turn left, then step. **Right:** turn right, then step.

Roles

Role	Job	Vocabulary
Expert	Walks to the goal	Demonstrator
Sensor	Walks beside whoever is walking and calls out F, L and R at every step	Observation
Scribe	Writes one row per step on the worksheet	One (observation, action) pair
Policy builders (2–4)	Tally the rows and write the lookup table	Training
Robot	Walks the room using only the table	The learned policy
Referee	Counts steps and bumps	Evaluation

Round 1: Collect demonstrations (10 min)

1. The Expert walks from a start to the goal. At each step the **Sensor** calls the reading, the **Expert** acts, and the **Scribe** fills in a row.
2. Do 3 or 4 walks from different starts, one worksheet page per walk.

Ask: The Expert can see the whole room, but the sheet records only 3 bits. Is that fair? (Partial observability. Save their answers for the discussion.)

Round 2: Build the policy (10 min)

1. Policy builders use the **Tally & Policy sheet**. For every row on every worksheet, add a tally mark in the line for that reading and the action taken.
2. For each reading, pick the action with the most tallies. Write it in the “Policy does” column. That table is the trained model.
3. Conflicts will appear: a reading with tallies for both left and right. The builders must choose (most tallies, or most recent). Nobody can “average” a left and a right turn into something sensible. (Multimodality.)

Round 3: Run the policy (10 min)

The Robot starts at a spot. Loop, with **no coaching from the Expert**: Sensor calls the reading → Robot finds the line in the table → Robot does that action. The Referee counts steps and bumps. Try a start the Expert used, and a new one.

Name each failure when it happens:

- **A reading with no tallies.** No card covered it. The Robot has to guess, or copy the most similar reading (nearest neighbor, as in the KNN activity). *Distribution shift.*
- **Spinning or looping.** A policy that only sees the current reading cannot know it has been here before. *Memoryless policy.*
- **Drifting into a corner the Expert never entered.** One slip leads to the next. *Compounding error.*

Round 4: Fix it with corrections (10 min)

1. Run the Robot again. The Expert follows along, and whenever the Robot is about to go wrong, says what they would do at that reading.
2. The Scribe records those rows in a **second ink color** on a new worksheet.
3. Policy builders add the tallies, rebuild the table, and run again. Compare steps and bumps before and after. *(DAgger.)*

Optional Round 5, recovery data: nudge the Expert off the usual line at the start (a shoulder tap, a random start square) and record the corrections. *(Noise injection.)*

Discussion questions

1. Which readings had conflicting rows, and why? Would the goal direction have resolved some?
2. The Expert could see a table beyond sensor range. How did that show up in the data as “mistakes”?
3. Why did the second-ink rows fix more than another full set of demonstrations would have?
4. What would remembering the last reading fix? Which failure would it not fix?
5. A looping policy has a sensible action for every reading. So what is wrong?

How it maps to the vocabulary

In the room	Term
The Expert’s walk	Demonstration, trajectory τ
One row on the worksheet	Pair (o, a)
All the worksheets together	Dataset D
The Policy “does” column	Trained policy π_θ
Tallies for two actions on one reading	Multimodality
A reading with no tallies	Out of distribution, distribution shift
Second-ink rows	DAgger corrections
The Expert sees more than 3 bits	Partial observability
Looping	A memoryless (Markovian) policy failing

Tips

- **Keep the sensors honest.** The Sensor reads the bits aloud so nobody quietly uses what they can see of the whole room.
- **Too hard?** Use only F (front blocked) with the actions forward and turn.
- **Too easy?** Add the goal direction, or run the Robot after fewer demonstrations to show where it fails.
- **Same activity on screen:** open `index.html` in the `bc_gridworld` folder. It has the same sensors, actions, cards, table, and correction round, so students can repeat it at home.